Solution 1 Sequence of Events

Problem 1.1 — Let $(A_n : n \in \mathbb{N})$ be a sequence of events, with $\mathbb{P}[A_n] = 1/n^2$ for all n. Set $X_n = n^2 1_{A_n} - 1$ and set $\overline{X}_n = (X_1 + \cdots + X_n)/n$. Show that $\mathbb{E}[\overline{X}_n] = 0$ for all n, but $\overline{X}_n \to -1$ almost surely as $n \to \infty$.

For any particular n, by linearity,

$$\begin{aligned}\mathbb{E}[\overline{X}_n] &= \mathbb{E}\left[\frac{X_1 + \cdots + X_n}{n}\right] \\ &= \frac{\mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n]}{n} \\ &= 0\end{aligned}$$

By, say upper-bounding a suffix by $\int_1^\infty \frac{1}{x^2}$, we see that

$$\sum_n \mathbb{P}[X_n] = \sum_n \frac{1}{n^2} < \infty$$

Thus, by Borel-Cantelli, A_n occurs finitely often almost surely; in this case, suppose that A_n never occurs for $n \geq N$. Then $X_n = -1$ for all $n \geq N$, so clearly $\overline{X}_n \to -1$.

Solution 2 Zeta Function

Problem 2.1 — The zeta function is defined for $s > 1$ by $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$. Let X and Y be independent random variables with $\mathbb{P}[X = n] = \mathbb{P}[Y = n] = n^{-s}/\zeta(s)$. Write A_n for the event that n divides X. Show that the events $(A_p : p \text{ prime})$ are independent and deduce Euler's formula

$$\frac{1}{\zeta(s)} = \prod_p \left(1 - \frac{1}{p^s}\right).$$

Claim — $\mathbb{P}[A_n] = \frac{1}{n^s}$ for all $n \in \mathbb{N}$.

Proof. Fix any $p \in \mathbb{N}$. A_p is the disjoint union of a countable number of events called $X = np$ for all $n \in \mathbb{N}$; as such, it is literally

$$\begin{aligned}
\mathbb{P}[A_p] &= \sum_{n \geq 1} \mathbb{P}[X = np] \\
&= \frac{1}{\zeta(s)} \sum_{n \geq 1} \frac{1}{(np)^s} \\
&= \frac{1}{p^s} \frac{1}{\zeta(s)} \sum_{n \geq 1} \frac{1}{n^s} \\
&= \frac{1}{p^s}
\end{aligned}$$

□

We sort the primes $p_1 < p_2 < \cdots$.

Claim — Given X as above, define the random variables P_i as the largest factor of p_i dividing X (these are measurable functions from $\mathbb{N}$ to $\mathbb{Z}_{\geq 0}$ because integers literally have discrete measure). Then, the random variables P_i are independent.

Proof. Fix a sequence of positive integers n_i; I show that

$$A_{p_i^{n_i}} = \{P_i \geq n_i\}$$

are independent.

Fix a finite set of distinct primes $\{p_i : i \in I\}$. Then,

$$A_M = \bigcap_{i \in I} A_{p_i^{n_i}}$$

where

$$M = \prod_{i \in I} p_i^{n_i}$$

Observe that

$$\mathbb{P}[A_M] = \frac{1}{M^s} = \prod_i \frac{1}{p_i^s} = \prod_i \mathbb{P}[A_{p_i}]$$

Note that A_{p^n} implies A_{p^m} for $m \leq n$. Thus, the above calculation proves the claim. □

Now, we want to compute $\mathbb{P}(X = 1)$. By elementary number theory, we know $\{X = 1\} = \bigcap\{P_i = 0\}$. Measures are continuous from above, thus

$$\begin{aligned}\mathbb{P}(X = 1) &= \lim_{m\to\infty} \mathbb{P}(P_i = 0 \text{ for all } i \le m) \\ &= \lim_{m\to\infty} \prod_{i=1}^{m} (1 - \mathbb{P}(A_i)) \\ &= \prod_{i=1}^{\infty} \left(1 - \frac{1}{p_i^s}\right)\end{aligned}$$

as desired.

Problem 2.2 — Show also that $\mathbb{P}[X \text{ is square free}] = 1/\zeta(2s)$. Write H for the highest common factor of X and Y. Show finally that $\mathbb{P}[H = n] = n^{-2s}/\zeta(2s)$.

Observe that the A_{p^2} are independent for all prime p by the same argument as before. Then, the probability that you are squarefree is precisely the countable intersection of the events $A_{p^2}^c$, which (by identical reasoning) is

$$\prod_p \left(1 - \frac{1}{(p^2)^s}\right) = \prod_p \left(1 - \frac{1}{p^{2s}}\right) = \frac{1}{\zeta(2s)}$$

Finally, we have two independent random variables X and Y, which induce prime-factorization random variables P_i and Q_i. All of $\{P_i\} \cup \{Q_i\}$ are mutually independent random variables, because P_i and Q_i are functions of X and Y, which are independent.

We've got to figure out the probability that $\gcd(X, Y) = n$. Let

$$n = \prod_{i=1}^{k} p_i^{n_i}$$

be the prime factorization of n. Then, we require $\min(P_i, Q_i) = n_i$. We now do the standard trick; this is a countable intersection of events called "$\min(P_i, Q_i) = n_i$ for all $i \le m$" which are all independent, so we can go directly to the limit.

Alright! So what's $\mathbb{P}(\min(P_i, Q_i) = n_i)$? Well, this is

$$\begin{aligned}\mathbb{P}(P_i = n_i, Q_i \ge n_i) \sqcup \mathbb{P}(P_i > n_i, Q_i = n_i) &= \mathbb{P}(P_i = n_i)\mathbb{P}(Q_i \ge n_i) + \mathbb{P}(P_i > n_i)\mathbb{P}(Q_i = n_i) \\ &= \left(\frac{1}{p_i^{n_i}} - \frac{1}{p_i^{n_i+1}}\right) \cdot \frac{1}{p_i^{n_i}} + \frac{1}{p_i^{n_i+1}} \cdot \left(\frac{1}{p_i^{n_i}} - \frac{1}{p_i^{n_i+1}}\right) \\ &= \left(\frac{1}{p_i^{n_i}} - \frac{1}{p_i^{n_i+1}}\right)\left(\frac{1}{p_i^{n_i}} + \frac{1}{p_i^{n_i+1}}\right) \\ &= \frac{1}{p_i^{2n_i}}\left(1 - \frac{1}{p_i^2}\right)\end{aligned}$$

Thus, the infinite product is

$$\begin{aligned}\prod_{i=1}^{\infty} \frac{1}{p_i^{2n_i}}\left(1 - \frac{1}{p_i^2}\right) &= \prod_{i=1}^{\infty} \frac{1}{p_i^{2n_i}} \prod_{i=1}^{\infty}\left(1 - \frac{1}{p_i^2}\right) \\ &= \frac{1}{n^{2s}} \prod_{i=1}^{\infty}\left(1 - \frac{1}{p_i^2}\right) \\ &= \frac{1}{n^{2s}} \frac{1}{\zeta(2s)}\end{aligned}$$

(I was allowed to split the first infinite product because all of the terms are < 1).

Solution 3 Cantor Set

Problem 3.1 — Let C_n denote the nth approximation to the Cantor set C: thus $C_0 = [0,1]$, $C_1 = [0,1/3] \cup [2/3,1]$, $C_2 = [0,1/9] \cup [2/9,1/3] \cup [2/3,7/9] \cup [8/9,1]$, etc. and $C = \bigcap_n C_n$. Denote by F_n the distribution function of a random variable uniformly distributed on C_n.

Show that C has Lebesgue measure 0.

Let $A_n \subset \{0,1,\ldots 3^n - 1\}$ consists of all numbers whose ternary expansion contains no 1s, in order; there are 2^n of these. Then,

$$C_n = \{[x \cdot 3^{-n}, (x+1) \cdot 3^{-n}] : x \in A_n\}$$

It's pretty clear that C_i has a Lebesgue measure of $\left(\frac{2}{3}\right)^i \to 0$, and $C_i \downarrow C$, thus C has measure 0.

Claim (Full Characterization of F_n) — By definition, $F_n(x)$ is the measure of $C_n \cap [0,x]$, divided by the measure of C_n. This is:

$$\begin{aligned}&\mu(C_n)^{-1}\left(\mu\left(C_n \cap \left([0,3^{-n}\cdot a] \cup [3^{-n}\cdot a, 3^{-n}\cdot a + 3^{-n}\cdot d]\right)\right)\right)\\ =&2^{-n}\left(\mu\left(\{[x,x+1] : x \in A_n\} \cap ([0,a] \cup [a,a+d])\right)\right)\end{aligned}$$

Let $b = |\{x : x \in A_n, x \le a-1\}|$ be the measure of the first intersection. Then, there's two cases

- If $a \notin A_n$, this is $2^{-n}b$.
- If $a \in A_n$, this is $2^{-n}(b+d)$

where $a = \lfloor 3^n x \rfloor$ and $d = \{3^n x\}$.

Problem 3.2 — For all $x \in [0,1]$, the limit $F(x) = \lim_{n\to\infty} F_n(x)$ exists.

Okay, fine! I claim that $|F_{n+1} - F_n| \le 2^{1-n}$ uniformly. For all x, let's track the progress of x through the above steps.

- Let $a_n = \lfloor 3^n x \rfloor, d_n = \{3^n x\}$, $a_{n+1} = \lfloor 3^{n+1} x \rfloor$, $d_{n+1} = \{3^{n+1}x\}$.
- Consider
$$b_{n+1} = |\{x : x \in A_{n+1}, x < a_{n+1}\}|$$
We know $3a_n \le a_{n+1} \le 3a_n + 2$, thus this set contains
$$\{3x : x \in A_n, x < a_n\} \cup \{3x+2 : x \in A_n, x < a_n\}$$
and is a subset of
$$\{3x : x \in A_n, x < a_n + 1\} \cup \{3x+2 : x \in A_n, x < a_n + 1\}$$
In particular, $2b_n \le b_{n+1} \le 2b_n + 2$.
- We also perform the idiotic bounds $0 \le d_{n+1}, d_n \le 1$.
- Okay. Now you can just subtract.

This probably kills the remaining parts. For part (ii), we just showed that

$$|F_n(x) - F_{n+1}(x)| < 2^{-n}$$

for all x, so the limit exists.

Problem 3.3 — The function F is continuous on $[0, 1]$, with $F(0) = 0$ and $F(1) = 1$.

We just showed that F_n was uniformly Cauchy convergent, so it is also uniformly convergent. A uniformly convergent series of continuous functions is continuous, so F is continuous.

Also, $F_n(0)$ is always 0, and $F_n(1)$ is always 1, so it's not deep.

Problem 3.4 — For almost all $x \in [0, 1]$, F is differentiable at x with $F'(x) = 0$.

For all $x \notin C$, there exists some i such that $x \notin C_i$. Of course, if $x \notin C_i$, then it is $\notin C_j$ for all $j \geq i$. Then, F_j is locally constant at x, for all $j \geq i$. thus F too will be locally constant at x, and has derivative 0.

Solution 4 Normal Distribution

Problem 4.1 — Show that, for all non-negative measurable functions f on $[0, \infty)$, the function $(x, y) \mapsto f(|(x, y)|)$ is measurable on $\mathbb{R}^2$.

$\phi : (x, y) \to |(x, y)|$ is measurable since it is continuous, and composition of measurable functions is measurable, thus $g = f \circ \phi$ is measurable.

Problem 4.2 — Show (without using the Jacobian formula)

$$\int_{\mathbb{R}^2} f(|(x, y)|)dxdy = 2\pi \int_0^\infty rf(r)dr.$$

Now, we're going to prove the statement for indicator functions of the form $f = 1_{[0,r]}$. Then, the LHS is

$$\begin{aligned}\int_{\mathbb{R}^2} 1_{[0,r]}(|(x, y)|)dxdy &= \int_{\mathbb{R}^2} 1_{B_r(0)}((x, y))dxdy \\ &= \pi r^2\end{aligned}$$

(This is true because this is just a Riemann integral, which coincides with the Lebesgue integral). The RHS is

$$\begin{aligned}2\pi \int_0^\infty rf(r)dr &= 2\pi \int_0^r rdr \\ &= \pi r^2\end{aligned}$$

as desired.

Let $g_n = f_n \circ \phi$. Then, $g_n \uparrow g$ are a series of simple functions as well. By the above results,

$$\int_{\mathbb{R}^2} g_n dxdy = 2\pi \int_0^\infty rg_n(r)dr$$

thus by the monotone convergence theorem, we conclude. This immediately implies the same result for arbitrary indicator functions 1_A, since $[0, r]$ is a π-system which generates the Borel σ-algebra on $[0, \infty)$.

By linearity, this in turn implies the same result for arbitrary simple functions.

Now, there exists some sequence of simple functions f_n such that $f_n \uparrow f$. Let $g_n = f_n \circ \phi$. Then, $g_n \uparrow g$ are a series of simple functions as well (a simple function is a finite linear combination of indicators on measurable sets, each of which is pulled back through ϕ to a measurable set).

By the above results,

$$\int_{\mathbb{R}^2} g_n dxdy = 2\pi \int_0^\infty rg_n(r)dr$$

thus by the monotone convergence theorem, we conclude.

Problem 4.3 — Hence show that $(2\pi)^{-1/2}e^{-x^2/2}$ is a probability measure.

We simply plug $f(x) = (2\pi)^{-1}e^{-x^2/2}$ into this formula. The RHS is

$$\int_0^\infty re^{-r^2/2}dr = \int_0^\infty e^{-u}du = 1$$

The LHS is

$$\begin{aligned}\int_{\mathbb{R}^2} f(|(x,y)|)dxdy &= \int_{\mathbb{R}^2} (2\pi)^{-1} e^{-\frac{x^2+y^2}{2}} dxdy \\ &= \left(\int_{\mathbb{R}} (2\pi)^{-1/2} e^{-\frac{x^2}{2}} dx\right)\left(\int_{\mathbb{R}} (2\pi)^{-1/2} e^{-\frac{y^2}{2}} dy\right) \\ &= \left(\int_{\mathbb{R}} (2\pi)^{-1/2} e^{-\frac{x^2}{2}} dx\right)^2\end{aligned}$$

Thus $\int_{\mathbb{R}} (2\pi)^{-1/2} e^{-\frac{x^2}{2}} dx = 1$, as desired. (It clearly can't be -1).

Solution 5 Random Variables

Problem 5.1 — Let X be a non-negative integer-valued random variable. Show that

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} \mathbb{P}[X \geq n].$$

Consider the sequence of simple functions

$$X_i = \sum_{n=1}^{i} 1_{\{X \geq n\}}$$

Then, $X_i \uparrow X$, thus $\mu(X_i) \to \mu(X)$. In particular,

$$\begin{aligned}
\mathbb{E}[X] &= \lim_{i\to\infty} \mathbb{E}[X_i] \\
&= \lim_{i\to\infty} \sum_{n=1}^{i} \mathbb{P}[X \geq n] \\
&= \sum_{n=1}^{\infty} \mathbb{P}[X \geq n]
\end{aligned}$$

Problem 5.2 — Deduce that, if $\mathbb{E}[X] = \infty$ and $X_1, X_2, \ldots$ is a sequence of independent random variables with the same distribution as X, then, almost surely, $\limsup_{n\to\infty}(X_n/n) \geq 1$, and moreover $\limsup_{n\to\infty}(X_n/n) = \infty$.

Pick any $k \in \mathbb{N}$.

$$\begin{aligned}
\infty &= \mathbb{E}[X] \\
&= \sum_{n=1}^{\infty} \mathbb{P}[X \geq n] \\
&= \sum_{n=1}^{\infty} \mathbb{P}[X \geq kn] \\
&= \sum_{n=1}^{\infty} \mathbb{P}[X_n \geq kn]
\end{aligned}$$

where the third equality is because $\mathbb{P}[X \geq n]$ is non-increasing in n, and thus e.g.

$$\mathbb{P}[X \geq kn] + \cdots + \mathbb{P}[X \geq kn + (k-1)] \leq k\mathbb{P}[X \geq kn]$$

but a sum is infinite iff all its tails are infinite, and we don't care about the multiplicative factor of k.

Since the X_n are independent, the second Borel-Cantelli lemma implies that $\mathbb{P}[X_n \geq kn \text{ i.o.}] = 1$. In particular, $\frac{X_n}{n} \geq k$ infinitely often, thus $\limsup_{n\to\infty} \frac{X_n}{n} \geq k$. But this is true for all k, so $\limsup_{n\to\infty} \frac{X_n}{n} = \infty$.

Problem 5.3 — Now suppose that $Y_1, Y_2, \ldots$ is any sequence of independent identically distributed random variables with $\mathbb{E}[Y_1] = \infty$. Show that, almost surely, $\limsup_{n\to\infty}(|Y_n|/n) = \infty$.

Let's let $Z_i = \lceil |Y_i| \rceil$. Note $\mathbb{E}[\lceil |Y_i| \rceil] \geq \mathbb{E}[Y_n] = \infty$, and $\lceil |Y_i| \rceil$ is non-negative, thus by the previous problem, for all $k \in \mathbb{N}$, $\limsup_{n\to\infty} \frac{Z_n}{n} \geq k$ almost surely.

Now, $|Y_n|$ differs from $\lceil |Y_n| \rceil$ by at most 1, thus $|Y_n| \geq nk - 1 \geq n(k-1)$ infinitely often. This holds for all k, so $\limsup_{n\to\infty} \frac{|Y_n|}{n} = \infty$.

Problem 5.4 — Show $\limsup_{n\to\infty}(|Y_1 + \cdots + Y_n|/n) = \infty$.

Suppose FSOC that $\limsup_{n\to\infty}(|Y_1 + \cdots + Y_n|/n) = k < \infty$. Then, there exists $N > 100$ such that for all $n > N$,

$$-nk \leq Y_1 + \cdots + Y_n \leq nk, \qquad -(n+1)k \leq Y_1 + \cdots + Y_{n+1} \leq (n+1)k$$

In particular, $|Y_{n+1}| \leq (2n+1)k$ for all $n > N$.

However, we just showed $|Y_n| \geq 100kn$ infinitely often, so this is a contradiction.

Solution 6 Bloody Integrals

Problem 6.1 — Show that as $n \to \infty$,

$$\int_0^\infty \frac{\sin(e^x)}{1+nx^2} dx \to 0$$

Consider

$$f_n(x) = \frac{\sin(e^x)}{1+nx^2}$$

Then $f_n \to f$ almost everywhere (in fact, for all $x > 0$). In addition, $|f(x)| \leq \frac{1}{1+x^2}$ for all n, x. Note that g is integrable, since its antiderivative is $\arctan(x)$ or something, and integrates to $\pi/2$.

Thus, the Dominated Convergence Theorem applies, showing

$$\mu(f_n) \to \mu(f) = 0$$

Problem 6.2 — Show that as $n \to \infty$,

$$\int_1^0 \frac{n\cos(x)}{1+n^2x^{3/2}} dx \to 0.$$

Consider

$$f_n(x) = \frac{n\cos(x)}{1+n^2x^{3/2}}$$

Then $f_n \to f$ almost everywhere (in fact, for all $x > 0$). We can bound this by

$$\frac{|n\cos x|}{1+n^2x^{3/2}} = \frac{|\cos x|}{n^{-1}+nx^{3/2}} \leq \frac{1}{2x^{3/4}}$$

where we used AM-GM in the denominator. However,

$$\int_0^1 \frac{x^{-3/4}}{2} dx = 2x^{1/4}\Big|_0^1 = 2 < \infty$$

Thus, the Dominated Convergence Theorem applies, showing

$$\mu(f_n) \to \mu(f) = 0$$

Solution 7 Pushforward Measure

Problem 7.1 — Let (E, E) and (G, G) be measurable spaces and let $f : E \to G$ be a measurable function. Given a measure μ on (E, E), consider the image measure $\nu = \mu \circ f^{-1}$ on (G, G). Show that $\nu(g) = \mu(g \circ f)$ for all non-negative measurable functions g on G.

Start with indicator functions. If $g = 1_A$ for some $A \in G$, then

$$\begin{aligned}\nu(g) &= \nu(1_A) \\ &= \mu(f^{-1}(A)) \\ &= \mu(g \circ f)\end{aligned}$$

By linearity, this holds for all simple functions.

Then, if $g_n \uparrow g$ is a sequence of simple functions, then $\nu(g_n) \uparrow \nu(g)$ are a sequence of simple functions, and $\mu(g_n \circ f) \uparrow \mu(g \circ f)$ are a sequence of simple functions. Thus, by the monotone convergence theorem,

$$\nu(g) = \lim_{n\to\infty} \nu(g_n) = \lim_{n\to\infty} \mu(g_n \circ f) = \mu(g \circ f)$$

Solution 8 Independence

Problem 8.1 — Let $X_1, \ldots, X_n$ be random variables with density functions $f_1, \ldots, f_n$ respectively. Suppose that the $\mathbb{R}^n$-valued random variable $X = (X_1, \ldots, X_n)$ also has a density function f. Show that $X_1, \ldots, X_n$ are independent if and only if

$$f(x_1, \ldots, x_n) = f(x_1) \cdots f(x_n) \quad \text{a.e.}$$

For the backwards direction, we simply integrate! Pick any tuple $(y_1, \ldots, y_n)$. We know that $f(x_1, \ldots, x_n) = f(x_1) \cdots f(x_n)$ almost everywhere, thus

$$\begin{aligned}
\mathbb{P}[X_1 \le y_1, \ldots, X_n \le y_n] &= \int_{x_i \le y_i} f(x_1, \ldots, x_n) dx_1 \cdots dx_n \\
&= \int_{\mathbb{R}^n} f(x_1) \cdots f(x_n) dx_1 \cdots dx_n \qquad = \prod_{i=1}^n \int_{x_i \le y_i} f(x_i) dx_i \\
&= \prod_{i=1}^n \mathbb{P}[X_i \le y_i]
\end{aligned}$$

showing independence.

In the other direction, we already know independence, and that density functions exist *at all*. So we just want to establish that $f(x_1, \ldots, x_n) = f(x_1) \cdots f(x_n)$ is in fact a density function for X, at which point it is the unique density function up to measure-0 sets.

Idea (Density functions are unique) — Pick ground space $(E, \mathcal{E}, \mu)$. Suppose we have two density functions $f : E \to \mathbb{R}$ and $f' : E \to \mathbb{R}$ of a measure which differ on a positive-measure set (in μ).

This positive-measure set is the countable union of sets of the form $f^{-1}((a, \infty)) \cup f'^{-1}((-\infty, a))$ for $a \in \mathbb{Q}$. Thus at least one such set has nonzero measure; suppose it is $A = f^{-1}((a, \infty)) \cup f'^{-1}((-\infty, a))$.

Then, $\int_A f \mathrm{d}\mu > a\mu(E)$ and $\int_A f'(x) \mathrm{d}\mu < a\mu(E)$. But these are both supposed to be equal to $\nu(A)$, which is a contradiction.

Well okay, this is not deep. We're going to start with boxes and work up from there; consider any box $A = A_1 \times \cdots \times A_n$ where each dimension is measurable. Then,

$$\begin{aligned}
\int_{A_1 \times \cdots \times A_n} f(x_1, \ldots, x_n) dx_1 \cdots dx_n &= \int_{A_1 \times \cdots \times A_n} f(x_1) \cdots f(x_n) dx_1 \cdots dx_n \\
&= \prod_{i=1}^n \int_{A_i} f(x_i) dx_i \\
&= \prod_{i=1}^n \mathbb{P}[X_i \in A_i] \\
&= \mathbb{P}[X \in A]
\end{aligned}$$

Now, the collection of measurable sets B such that

$$\int_B f(x_1, \ldots, x_n) dx_1 \cdots dx_n = \mathbb{P}[X \in B]$$

forms a λ-system, because:

- E is a box, so the above holds.
- If $B_1 \subset B_2$, then

$$\begin{aligned}\int_{B_2 \setminus B_1} f(x_1, \ldots, x_n) dx_1 \cdots dx_n &= \int_{B_2} f(x_1, \ldots, x_n) dx_1 \cdots dx_n - \int_{B_1} f(x_1, \ldots, x_n) dx_1 \cdots dx_n \\ &= \int_{B_2} f(x_1) \cdots f(x_n) dx_1 \cdots dx_n - \int_{B_1} f(x_1) \cdots f(x_n) dx_1 \cdots dx_n \\ &= \int_{B_2 \setminus B_1} f(x_1, \ldots, x_n) dx_1 \cdots dx_n\end{aligned}$$

- If $B_i \uparrow B$, then by the monotone convergence theorem,

$$\begin{aligned}\int_B f(x_1, \ldots, x_n) dx_1 \cdots dx_n &= \lim_{i \to \infty} \int_{B_i} f(x_1, \ldots, x_n) dx_1 \cdots dx_n \\ &= \lim_{i \to \infty} \int_{B_i} f(x_1) \cdots f(x_n) dx_1 \cdots dx_n \\ &= \int_B f(x_1) \cdots f(x_n) dx_1 \cdots dx_n\end{aligned}$$

It also contains all boxes, thus by the π-λ theorem, it contains all measurable sets (this is how the product measure is defined in the first place).